⌈Def. Define

$$f:[0,1]\to\mathbb{R}: x\mapsto \begin{cases} \frac{1}{p} & \text{if } x\in\mathbb{Q} \text{ with } x=\frac{q}{p}, \text{ irreducible form,} \\ 0 & \text{otherwise.} \end{cases}$$

Then,

1). f is continuous at every irrational numbers, and discontinuous at every rational numbers.

2) f is Riemann-integrable.

Proof. observe that for each $n\in\mathbb{N}$, the set

$$S_n=\{x\in[0,1] \mid f(x)\geq\tfrac{1}{n}\}=f^{-1}[[\tfrac{1}{n},\infty)]=\{\tfrac{q}{p}\in[0,1] \mid \gcd(p,q)=1,\ 1\leq p\leq n\}.$$

is finite set,

Let $x_0\in[0,1]$ be an irrational number.

Given $\varepsilon>0$, there is $N\in\mathbb{N}$ s.t $\frac{1}{N}<\varepsilon$. Since the S_N is finite, we can take $\delta>0$ s.t $\left(\delta = \min\{|x_0-y| \mid y\in S_N\}\right)$

$$x\in(x_0-\delta,x_0+\delta)\Rightarrow |f(x_0)-f(x)|=f(x)\leq\frac{1}{N}<\varepsilon.$$

so, f is continuous at any irrational. And, for rational $\frac{q}{p}\in[0,1]$,

for any $\delta>0$, there is an irrational $x\in(\frac{q}{p}-\delta,\frac{q}{p}+\delta)$ s.t $|\frac{1}{p}-f(x)|=\frac{1}{p}>\frac{1}{2p}$.

So it is not continuous at rational. ⌋ 1).

Given $\varepsilon>0$, let $N\in\mathbb{N}$ be a number such that $\frac{1}{N}<\frac{\varepsilon}{2}$.

Denote that $S_N=\{x_1,\cdots,x_k\}$. Since this S_N is finite, put $d>0$ such that

$$d<\min\{|x_i-x_j| \mid i\neq j\} \quad \text{and} \quad d\cdot k<\frac{\varepsilon}{2}.$$

Now, construct a partition on $[0,1]$ as

$$P=\{0,1,x_1-\tfrac{d}{2},x_1+\tfrac{d}{2},\cdots,x_k-\tfrac{d}{2},x_k+\tfrac{d}{2}\}\cap[0,1].$$

Then, by the density of irrationals, the lower sum $L(f,P)=0$, and

$$U(f,P)\leq 1\cdot\frac{1}{N}+1\cdot d\cdot k$$

Now, $\overline{\int} f-\underline{\int} f\leq U(f,P)-L(f,P)\leq\frac{1}{N}+dk<\frac{\varepsilon}{2}+\frac{\varepsilon}{2}$

gives the result, by the integrable criterion. ☒